THE KUBERT–TATE NORMAL FORM AND GOOD REDUCTION EVERYWHERE

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This worksheet is about two sections of [Sil09]: torsion points (§VIII.7), and minimal equations and good reduction (§VIII.8). Mazur's theorem lists the fifteen groups that occur as the torsion subgroup of an elliptic curve over $\mathbb{Q}$. The Kubert–Tate normal form puts an elliptic curve with a k-point P of order $m \geqslant 4$ in a family with two parameters, in which m is cut out by a polynomial condition on the parameters. These conditions define curves, the modular curves $Y_1(m)$, whose k-points parametrize such pairs (E, P). Problem 1.4 derives the normal form, and finds the conditions for $m = 4, 5, 6, 7, 9, 11, 12$. For $m = 11$ the curve has genus one. Problem 2.5 shows that every elliptic curve over $\mathbb{Q}$ has bad reduction at some prime. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A field k, and an elliptic curve E over k given by a Weierstrass equation
$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$
with coefficients in k, discriminant Δ and point at infinity O [Sil09, §III.1].
- A change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$, with $u \in k^\times$ and $r, s, t \in k$. It gives a Weierstrass equation for E in x', y', with discriminant $u^{-12}\Delta$. Any two Weierstrass equations for E are related by a change of variables [Sil09, Proposition III.3.1(b)].
- Coordinates $x(Q)$ and $y(Q)$ of a $\overline{k}$-point $Q \neq O$ of E.

How to use this. Allow about 50 minutes for Problem 1.4 and 20 for Problem 2.5. Magma is needed only in Problem 1.4(9). Everything else is by hand.

1. The Kubert–Tate normal form and Mazur's list

Problem 1.4 is [Sil09, Exercise 8.13]. Its item (1) is Exercise 8.13(a), the orders 6, 7, 9 and 12 are Exercise 8.13(b), and the order 11 is Exercise 8.13(c). The normal form, and the conditions for all the orders allowed by Theorem 1.1, are in [Kub76]. The elliptic curves with a k-point of order $m \geqslant 4$ are parametrized by the k-points of a modular curve [Sil09, Remark VIII.7.8].

Theorem 1.1 (Mazur [Sil09, Theorem VIII.7.5])**.** *Let E be an elliptic curve over $\mathbb{Q}$. Then $E(\mathbb{Q})_{\mathrm{tors}}$ is isomorphic to one of the fifteen groups*
$$\mathbb{Z}/N\mathbb{Z} \text{ with } 1 \leqslant N \leqslant 10 \text{ or } N = 12, \qquad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2N\mathbb{Z} \text{ with } 1 \leqslant N \leqslant 4.$$
Each of them occurs as $E(\mathbb{Q})_{\mathrm{tors}}$ for some elliptic curve E over $\mathbb{Q}$.

Proposition 1.2 (The group law [Sil09, Proposition III.2.2(a) and Algorithm III.2.3(a)])**.** *If a line in $\mathbb{P}^2$ meets E at the $\overline{k}$-points P_1, P_2, P_3, counted with multiplicity, then $P_1 + P_2 + P_3 = O$. The negative of a $\overline{k}$-point (x_0, y_0) is $(x_0, -y_0 - a_1x_0 - a_3)$.*

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Definition 1.3 (Kubert–Tate normal form)**.** A Weierstrass equation is in Kubert–Tate normal form if it is

$$y^2 + (1-c)xy - by = x^3 - bx^2 \tag{1.1}$$

for some $b, c \in k$.

Problem 1.4 (The Kubert–Tate normal form and Mazur's list)**.**
(1) Let $P \in E(k)$ be a k-point of order $m \geqslant 4$, possibly infinite. In three steps, show that a change of variables gives a Weierstrass equation for E in Kubert–Tate normal form, in which $P = (0,0)$. After each change of variables, write again x, y for the coordinates and $a_1, \ldots, a_6$ for the coefficients.
(a) Show that a change of variables with $u = 1$ and $s = 0$ moves P to $(0,0)$, and that then $a_6 = 0$.
(b) Suppose that $P = (0,0)$, so that $a_6 = 0$. Show that $a_3 \neq 0$, and that the change of variables with $u = 1$, $r = t = 0$ and $s = a_4/a_3$ makes $a_4 = 0$. Show that then the tangent line to E at P is $y = 0$.[a]
(c) Suppose that $P = (0,0)$ and $a_4 = a_6 = 0$, with $a_3 \neq 0$. Show that $a_2 \neq 0$, and that the change of variables with $u = a_3/a_2$ and $r = s = t = 0$ gives a Weierstrass equation in Kubert–Tate normal form. Find b and c.[b]
(2) Conversely, suppose that E is given by an equation in Kubert–Tate normal form. Let $P := (0,0) \in E(k)$, and let m be its order, possibly infinite. Show that $b \neq 0$, that $2P = (b, bc)$ and $3P = (c, b-c)$, and that $m \geqslant 4$.[c]
(3) With E and P as in (2), suppose that a change of variables gives another equation for E in Kubert–Tate normal form, with parameters b' and c', in which P is again $(0,0)$. Show that $b' = b$ and $c' = c$. So the pair (E, P) determines b and c.[d]
(4) With E, P and m as in (2), show that $m = 4$ if and only if $c = 0$, and $m = 5$ if and only if $b = c$.
(5) With E, P and m as in (2), find the condition on b and c for $m = 6$.
(6) With E, P and m as in (2), suppose that $m \geqslant 5$, so that $c \neq 0$, and put $d := b/c$. Show that[e]

$$4P = \big(d^2 - d,\ d^2(c - d + 1)\big).$$

(7) With $m \geqslant 5$ and d as in (6), show that $m = 7$ if and only if P, $2P$ and $4P$ are collinear, and that this holds if and only if $c = d^2 - d$. Deduce a one-parameter family of elliptic curves with a k-point of order 7.
(8) With d as in (6), suppose that $m \geqslant 6$, so that $d \neq 1$, and put $f := c/(d-1)$. Show that $m = 9$ if and only if $2P$, $3P$ and $4P$ are collinear, and that this holds if and only if

$$c^2 - (d-1)c - (d-1)^3 = 0, \quad \text{that is,} \quad d = f^2 - f + 1.$$

Deduce a one-parameter family of elliptic curves with a k-point of order 9.[f]
(9) With d and f as in (8), suppose that $m \geqslant 9$, so that $5P, \ldots, 8P$ are not O. The following code takes $b = cd$, works over the field $\mathbb{Q}(c,d)$ of rational functions, and factors the numerators of $x(4P) - x(8P)$ and $x(5P) - x(6P)$ after the substitution $c = f(d-1)$.

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K<c, d> := FunctionField(Rationals(), 2);
E := EllipticCurve([1 - c, -c*d, -c*d, 0, 0]);     // b = c*d
P := E![0, 0];
4*P;                        // (d^2 - d : c*d^2 - d^3 + d^2 : 1)
R<f, d> := PolynomialRing(Rationals(), 2);
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// the irreducible factors of the numerator of x(iP) - x(jP), with c = f(d - 1)
Factors := func<i, j | [g[1] : g in Factorization(
  Evaluate(Numerator((i*P)[1] - (j*P)[1]), [f*(d - 1), d]))]>;
Factors(4, 8);
// [ d, d - 1, f - 1, f^2 - f*d^2 - f*d + 3*d^2 - 3*d + 1 ]
Factors(5, 6);
// [ d - 1, f, f^3*d - 3*f^2*d + 4*f*d - f - d^2 ]
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(a) Show that $m = 12$ if and only if

$$f^2 - (d^2 + d)f + 3d^2 - 3d + 1 = 0.$$

Suppose that $\operatorname{char} k \neq 2$. Find a one-parameter family of elliptic curves with a k-point of order 12.[g]

(b) Show that $m = 11$ if and only if

$$d^2 - (f^3 - 3f^2 + 4f)d + f = 0.$$

Suppose that $\operatorname{char} k \notin \{2, 11\}$. Show that this curve is birational to the curve

$$w^2 = f(f^3 - 4f^2 + 8f - 4),$$

and that the latter has genus one.[h]

[a] *Hint:* the tangent line to E at P is $a_3 y = a_4 x$. If $a_3 = 0$, it passes through O.
[b] *Hint:* if $a_2 = 0$, the tangent line $y = 0$ meets E only at P.
[c] *Hint:* find the tangent line to E at P, and the line through P and $2P$. Use Proposition 1.2.
[d] *Hint:* the change of variables takes the $\overline{k}$-point $(x', y') = (0, 0)$ to $(x, y) = (r, t)$. For $r = t = 0$, compute the new coefficients $a'_1, \ldots, a'_4$.
[e] *Hint:* the line through P and $3P$ is $y = (d - 1)x$.
[f] *Hint:* the line through $2P$ and $3P$ has slope $(cd - d + 1)/(d - 1)$.
[g] *Hint:* as a polynomial in f, its discriminant is $(d - 1)^2(d^2 + 4d - 4)$. Parametrize the conic $(d + 2)^2 - w^2 = 8$ by $d + 2 - w = 2v$.
[h] *Hint:* complete the square in d. The discriminant of $f^3 - 4f^2 + 8f - 4$ is -176.

2. No good reduction everywhere

This section is [Sil09, Exercise 8.15], and follows its hint. A Weierstrass equation over $\mathbb{Z}$ with $\Delta = \pm 1$ would have good reduction at every prime, and congruences modulo 8 rule it out.

In the following definitions and results, E is an elliptic curve over $\mathbb{Q}$ and p is a prime number, with valuation v_p. Write $\mathbb{Z}_{(p)}$ for the ring of rational numbers whose denominators are prime to p.

Definition 2.1 (Minimal equations [Sil09, §VII.1])**.** A Weierstrass equation for E is minimal at p if its coefficients lie in $\mathbb{Z}_{(p)}$ and $v_p(\Delta)$ is as small as possible among such equations for E.

Proposition 2.2 (Good reduction [Sil09, Proposition VII.5.1(a)])**.** *The curve E has good reduction at p if and only if a Weierstrass equation for E minimal at p has $v_p(\Delta) = 0$.*

Corollary 2.3 (Global minimal equations [Sil09, Corollary VIII.8.3])**.** *The curve E has a Weierstrass equation with coefficients in $\mathbb{Z}$ that is minimal at every prime p.*

Proposition 2.4 (Weierstrass formulas [Sil09, §III.1])**.** *Put*

$$b_2 = a_1^2 + 4a_2, \qquad b_4 = 2a_4 + a_1 a_3, \qquad b_6 = a_3^2 + 4a_6,$$
$$b_8 = a_1^2 a_6 + 4a_2 a_6 - a_1 a_3 a_4 + a_2 a_3^2 - a_4^2, \qquad c_4 = b_2^2 - 24b_4, \qquad c_6 = -b_2^3 + 36 b_2 b_4 - 216 b_6.$$

Then $\Delta = -b_2^2 b_8 - 8b_4^3 - 27b_6^2 + 9b_2 b_4 b_6$, *and* $1728\Delta = c_4^3 - c_6^2$.

Problem 2.5 (No good reduction everywhere)**.**
Let E be an elliptic curve over $\mathbb{Q}$. Suppose, for a contradiction, that E has good reduction at every prime p.

(1) Show that E has a Weierstrass equation with coefficients in $\mathbb{Z}$ and $\Delta = \pm 1$. Fix such an equation, and put $\varepsilon := \Delta$.
(2) Show that a_1 is odd.[a]
(3) Show that $c_4 \equiv 1 \pmod 8$.
(4) Put $u := c_4 - 12\varepsilon$. Show that

$$c_6^2 = u\,(u^2 + 36\varepsilon u + 432),$$

and deduce that u is a positive square or three times a positive square.[b]
(5) Show that $u \equiv 5 \pmod 8$, and conclude that E has bad reduction at some prime p.

[a] *Hint:* if a_1 is even, show that $\Delta \equiv -27b_6^2 \pmod 8$.
[b] *Hint:* the integer u is odd, and every common divisor of the two factors divides $432 = 2^4 \cdot 3^3$.

References

[Kub76] D. S. Kubert, *Universal bounds on the torsion of elliptic curves*, Proc. London Math. Soc. (3) **33** (1976), no. 2, 193–237. ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑